Do Construction Spending and Jobs predict U.S Poverty Rates?

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Abstract

This project asks whether highway and street construction spending and the number of jobs predict the poverty rate in the US. We created a dataset for 2002-2024 using public data. We gathered highway and street construction spending, total nonfarm payroll jobs, and the Census Bureau's poverty rate estimate. Poverty rose from 12.1 in 2002 to a peak of 15.9% after the 2008 recession in 2011-2012. Then fell to 11.9% in 2020. Years with more jobs tend to have lower poverty with a correlation of -0.45, jobs alone explain about 19% of the variation in poverty. Construction spending has a weaker correlation with -0.21. That relationship depends only on 2023 and 2024. The two predictors are highly correlated with eachother .79.

Introduction

The poverty rate is one of the most widely used metrics for economic well-being. We also understand governments spend large amounts on infrastructure, and spending is defended with creating jobs and supporting economies. Does infrastructure construction however and the number of jobs predict poverty rates in the United States?

This analysis may benefit policymakers and the public, who want to know whether construction spending and job growth are connected to poverty. We are not proving cause and effect however. Instead, we describe how each variable overtime, measure how the predictors are related to poverty, and fit regression models to see how much variation in poverty is explained by the predictors.

Executive Summary

Jobs matter more than construction Jobs Correlation -0.43, COnstruction -0.23 **Poverty Follows Economy** It peaks at 15.9% in 2011-2012 after the great recession and its low in 2020 of 11.9%.

The predictors overlap. COnstruction and Jobs are correlatied at .79

Data

All of our Data comes from Federal Reserve Economic Data, a database ran by the Federal Reserve Bank of St. Louis. The other statistics are provided by the U.S. Census Bureau and the Bureau of Labor Statistics.

We downloaded four serieses as CSV files and our original files are in "data/raw/":

TLHWCON: Total highway and street construction spending (Census Bureau). Monthly, in millions of dollars **PAYEMS**: Total nonfarm payroll employment (Bureau of Labor Statistics). Monthly, in thousands

of people **PPAAUS00000A156NCEN**: Poverty rate for all ages, from the Small Area Income and Poverty Estimates (Census Bureau). Annual, in percent of people **CPIAUCSL**: Consumer Price Index for All Urban Consumers. (Bureau of Labor Statistics). Monthly, as an index

Cleaning and Processing All of our steps are seen in "11_DataCleaning_Processing.Rmd", using functions from "02_Functions.R"

1. Each file was read, and checked for errors and duplicates, missing values were counted. Poverty Series was missing: 1990, 1991, 1992, and 1994. CPI series is missing October 2025. Missing values were removed.
2. A year was kept only if all 12 months were present. Construction spending was summed up to get a yearly spending. Monthly jobs and CPI values were averaged for all 12 months. This is to keep it matching to our poverty rate
3. We adjusted Construction spending into 2024 dollars using the CPI for each year.
4. We matched each by year, and if one of the series was missing a year, we removed it. The dataseries thus starts in 2002 and ends in 2024.
5. A final recheck is done for duplicate years and missing or negative values and saved to "us_annual_2002_2024.csv"

Our final dataset has 23 observations, and the variables: 'year', 'construction_nominal_musd', 'construction_real_2024_musd', 'jobs_thousands', 'jobs_millions', 'cpi_index', and 'poverty_pct'. Our analysis checks 'poverty_pct', 'construction_real_2024_musd', and 'jobs_millions'

** Summary Statistics**

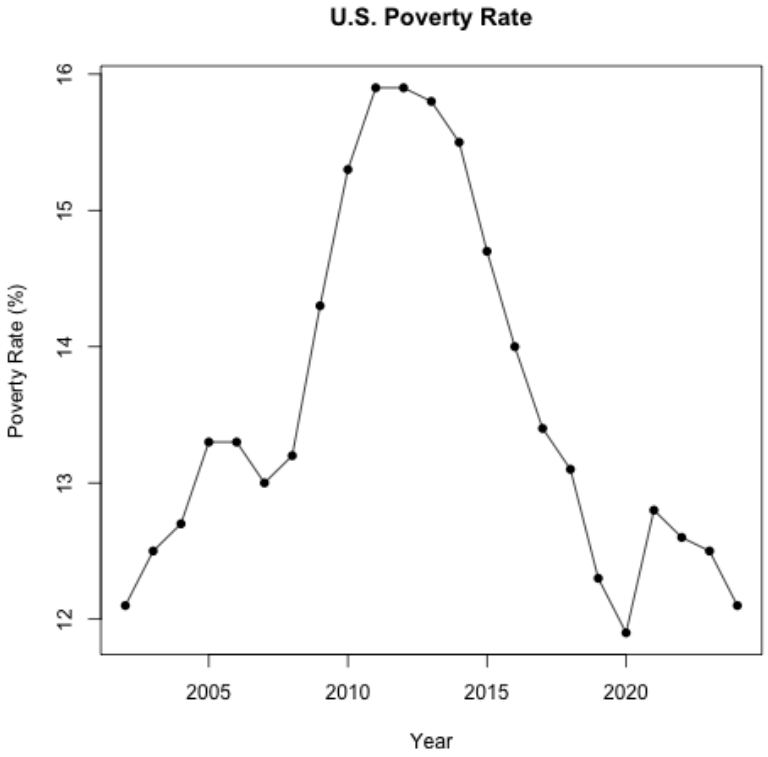
```
##      Poverty rate (%) Construction (billion 2024 USD) Jobs (millions)
## N               23.00                          23.00           23.00
## Mean            13.57                         116.54          140.37
## SD               1.33                          12.31            8.58
## Min             11.90                          97.35          130.33
## Max             15.90                         148.34          157.69
```

Visualization

Visualization Plans: We need to add histograms of each variable, box plots, and scatter plots of year to year and residual plots.

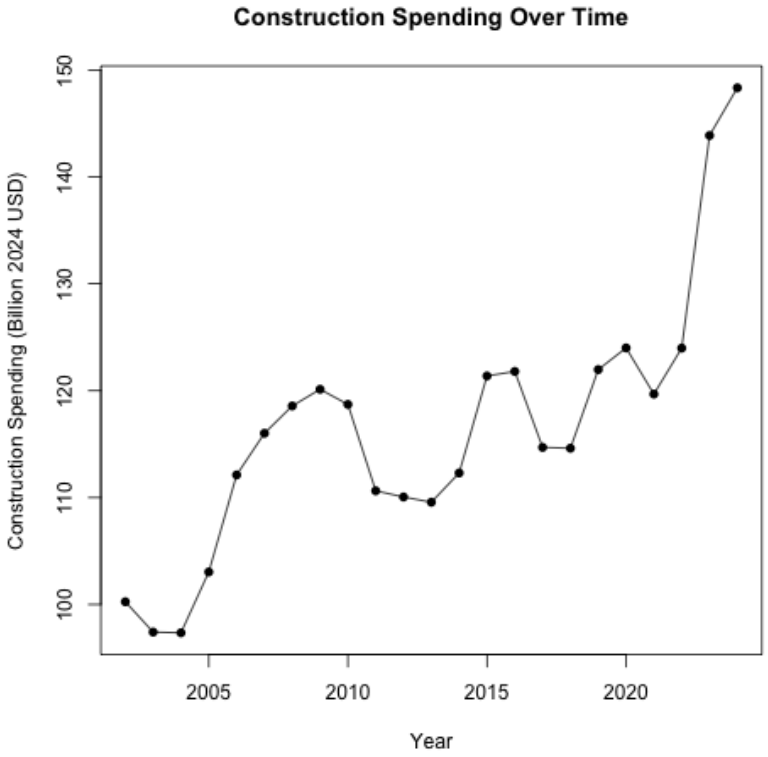
Poverty over time. Figure 1 shows that poverty did not move in one direction. It rose, then exploded after the 2008 Recession, to a peak of 15.9& in 2011 and 2012. Then declined to 11.9% in 2020.

```
knitr::include_graphics("figures/01_poverty_trend.png")
```

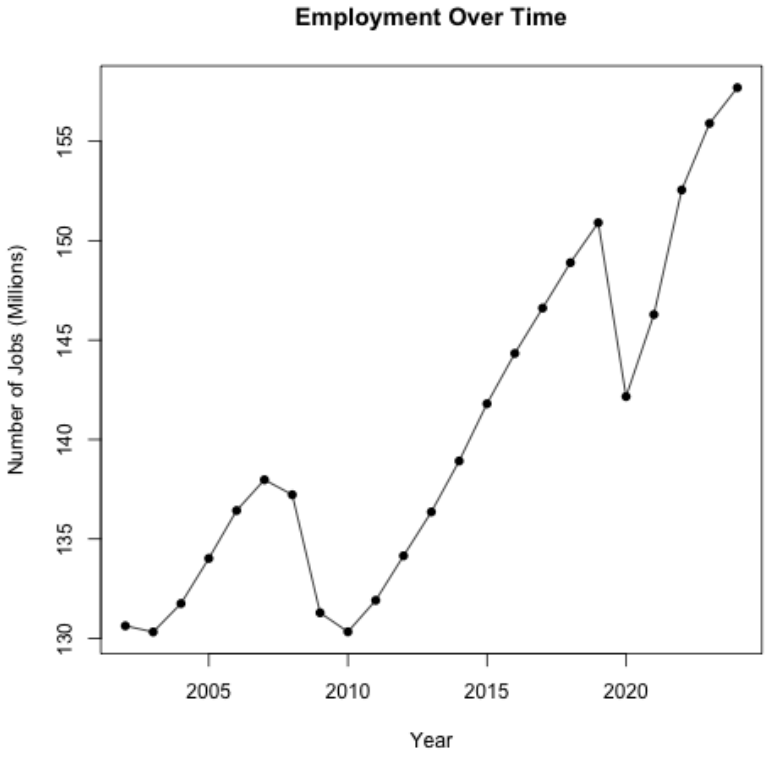


Construction spending and jobs overtime. Figure 2 shows that both predictors grew in different patterns. Jobs fell in the 2008 Recession, and again in 2020 during Covid-19, recovering after each drop. Real construction spending rose until the 2008 Recession, dipped in the 2010s, and then jumped sharply in 2023 and 2024. Comparing figure 1 and figure 2 shows two useful patterns. Poverty kept rising until 2011, even though jobs bottomed out at 2010, which suggests poverty responds to the job market but with a 1-2 year delay. Poverty reached its lowest value in 2020, the same year jobs dropped. Which suggest outside factors affect poverty, which we assume is Covid-19

```
knitr::include_graphics("figures/04_construction_trend.png")
```

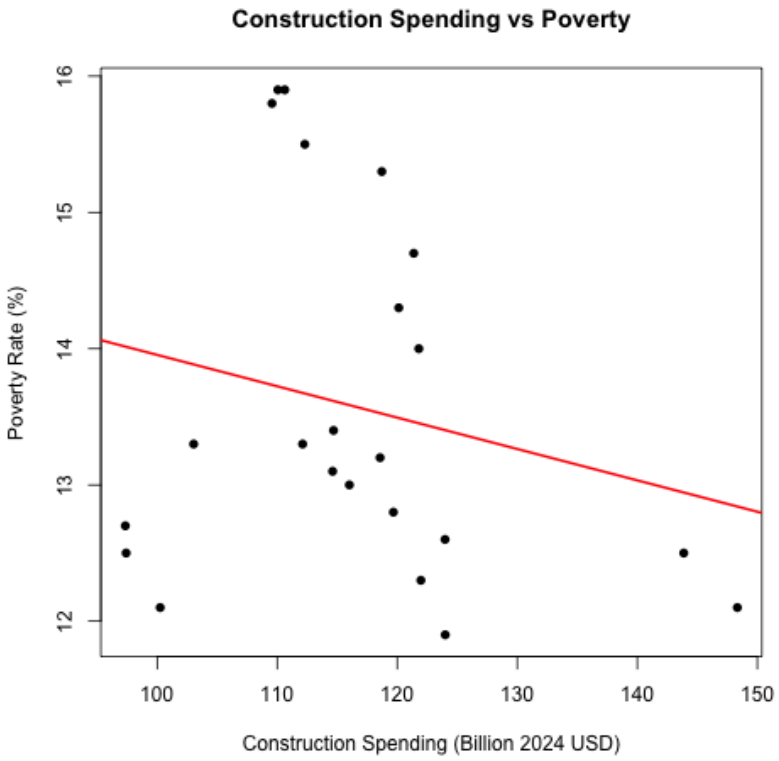



```
knitr::include_graphics("figures/05_jobs_trend.png")
```

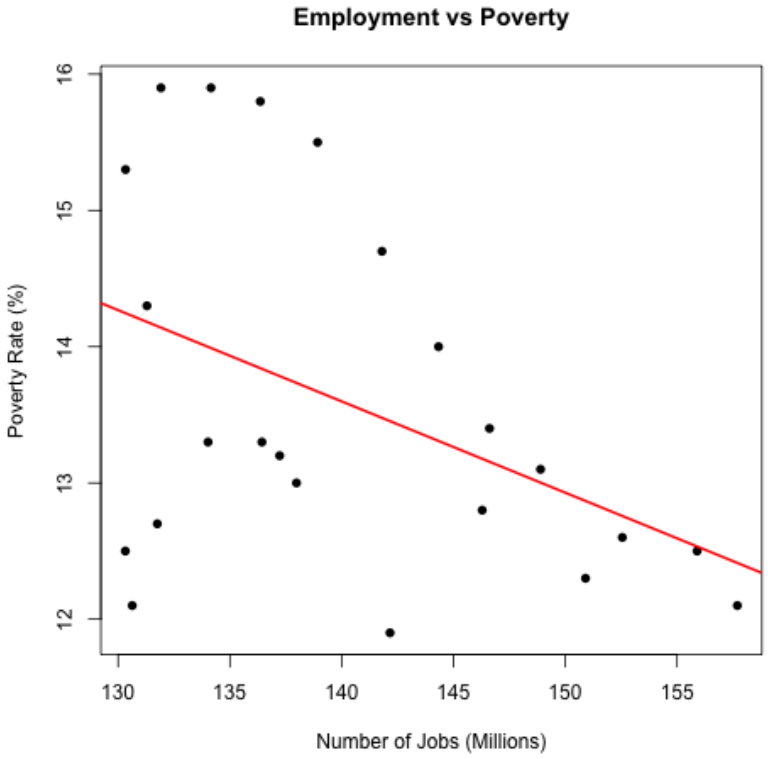


Relationships with poverty Figure 3 plots poverty against each predictor with a fitted regression line. Both of the lines slope down, but the points are scattered. The jobs scatterplot shows a clearer downward pattern however. The construction scatterplot, shows the two years with the highest spending, 2023 and 2024, sit far to the right of the other points and both have low poverty, so their influence on the line is large.

```
knitr::include_graphics("figures/02_construction_vs_poverty.png")
```




```
knitr::include_graphics("figures/03_jobs_vs_poverty.png")
```



Correlations:

```
##                              poverty_pct construction_real_2024_musd
## poverty_pct                        1.000                       -0.214
## construction_real_2024_musd       -0.214                        1.000
## jobs_millions                     -0.434                        0.791
##                              jobs_millions
## poverty_pct                         -0.434
## construction_real_2024_musd          0.791
## jobs_millions                        1.000
```

This correlation matrix confirms what the scatter plots are showing. Poverty has a weak negative correlation with construction of -0.214 and a moderate negative correlation with jobs -0.434, the two predicators are strongly correlated with eachother 0.791. This is important for the regression, since it is hard to tell which one is responsible for the changes in poverty.

Simple linear Regression

For one predictor x, and the response y, the regression line can be found with the formulas: b=r* sy/ sx, a = ybar- bxbar, R^2 = r^2

Where r is the correlation, sx and sy are the standard devations, and xbar and ybar are the means.

We made a function "simple_regression" to find this,

```
##                 predictor intercepts      slopes r_squareds
## 1 Construction (billions)   16.25819 -0.02303237 0.04568725
## 2         Jobs (millions)   22.98042 -0.06701280 0.18804213
```

Analysis Plan

1. Create a multiple linear regression
2. Check year to year changes
3. Check out delayed effects
4. See how years like 2020 change the correlation

We expect jobs to stay negatively related to poverty.